

Tutorial

1. Show that if $\Psi_1(x,t)$ and $\Psi_2(x,t)$ are 2 solutions of the Schrodinger equation, then $\Psi(x,t) = A\Psi_1 + B\Psi_2$ is also a solution of the Schrodinger equation.
2. If $\psi_n(x)$ is a solution of the TISE with energy E_n , then show that $\Psi(x,t) = \sum_n (A_n \psi_n(x) \exp(-iE_n t/\hbar))$ is a solution of the Schrodinger equation.
3. $\psi_n(x)$ is a solution of the TISE with energy E_n . We know that $\psi_n(x)$ is a stationary state. Show that $\Psi(x,t) = \sum_n (A_n \psi_n(x) \exp(-iE_n t/\hbar))$ is not a stationary state.
4. The Gaussian wave function is given by $\psi(x) = A \exp [-(x-\mu)^2 / 4\sigma^2]$. Calculate the normalization constant A .
5. A particle starts with wavefunction $\Psi(x,0) = (3/5)\psi_0(x) + (4/5)\psi_1(x)$, where $\psi_0(x)$ and $\psi_1(x)$ are stationary states. Find the probability density at some later time. What is the expectation value for the energy? Given, $\psi_0(x)$ and $\psi_1(x)$ are real and orthonormal.
6. For an infinite square well, determine the probability of finding the particle between $x = 0.25a$ and $x = 0.75a$ in the ground state and the 1st excited state.
7. A particle in infinite square well has initial wave function $\Psi(x, 0) = A x (a - x)$, for $0 \leq x \leq a$, for some constant A . Outside the well $\Psi(x,0) = 0$. Find $\Psi(x,t)$.